

30-Minute Student Task

Build Your Own Rule-Based Python Chatbot

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TASK	Rule-Based Conversational Chatbot

1. Objective

The objective is to build a simple rule-based conversational chatbot in Python. The chatbot continuously accepts user messages, processes them using conditional logic, responds to recognized messages, and exits when the user types **bye** or **exit**.

2. Python Program

```
# Rule-Based Python Chatbot
# Name: DODDI SRICHARAN RAO
# Reg No: 99230040099

print("Welcome to the Rule-Based Chatbot!")

name = input("What is your name? ").strip()
print("Hello", name + "! You can chat with me.")
print("Type 'bye' or 'exit' to end the conversation.")

while True:
    message = input("You: ").lower().strip()

    if message == "bye" or message == "exit":
        print("Bot: Goodbye", name + "! Have a great day!")
        break

    elif "hello" in message or "hi" in message:
        print("Bot: Hello", name + "! How can I help you?")

    elif "how are you" in message:
        print("Bot: I am doing well. Thanks for asking!")

    elif "your name" in message:
        print("Bot: I am a simple rule-based Python chatbot.")

    elif "python" in message:
        print("Bot: Python is a popular programming language used for many applications.")

    elif "weather" in message:
        print("Bot: I cannot check live weather, but I hope the weather is pleasant!")

    else:
        print("Bot: Sorry, I do not understand that message. Try asking about Python or saying hello.")
```

3. How the Program Works

Step	Description
1	The program asks the user for their name before the main conversation.

Step	Description
2	The while True loop keeps the chatbot running continuously.
3	input() receives each message from the user.
4	.lower().strip() converts the message to lowercase and removes extra spaces.
5	if / elif / else checks greetings, questions, topics, and unknown messages.
6	The in operator allows variations such as 'hello bot' and 'hey hello'.
7	break stops the loop when the user types 'bye' or 'exit'.

4. Requirement Checklist

Requirement	Implemented
Ask the user's name before the main conversation	Yes
Use while True:	Yes
Use input() to receive messages	Yes
Clean input using .lower().strip()	Yes
Recognize at least two greetings	Yes — hello and hi
Answer at least three questions/topics	Yes — Python, weather, name, and more
Use if / elif / else	Yes
Use break for bye or exit	Yes
Use the user's name in a response	Yes
Useful response for unknown messages using else	Yes
Use in operator for flexible greetings	Yes

5. Sample Test Run

```
Welcome to the Rule-Based Chatbot!
What is your name? Sricharan
Hello Sricharan! You can chat with me.
Type 'bye' or 'exit' to end the conversation.
```

```
You: hello
Bot: Hello Sricharan! How can I help you?
```

```
You: HELLO
Bot: Hello Sricharan! How can I help you?
```

```
You: hi
Bot: Hello Sricharan! How can I help you?
```

```
You: tell me about python
Bot: Python is a popular programming language used for many applications.
```

```
You: what is your name
Bot: I am a simple rule-based Python chatbot.
```

```
You: xyzabc
Bot: Sorry, I do not understand that message. Try asking about Python or saying hello.
```

```
You: bye
Bot: Goodbye Sricharan! Have a great day!
```

6. Conclusion

The chatbot successfully follows the required INPUT → PROCESS → RESPOND → REPEAT → EXIT flow. It uses Python input handling, string cleaning, conditional statements, the in operator, a continuous while loop, and break-based termination. The program also provides a useful fallback response when the message is not recognized.